

Sec 1.9 – Further Limit Calculations Using Algebra

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

Limits of Quotients:

In calculus we often encounter limits of the form $\lim_{x \rightarrow c} f(x) / g(x)$ where $f(x)$ and $g(x)$ are continuous.

There are three types of behavior for this type of limit:

- When $g(c) \neq 0$, the limit can be evaluated by substitution.
- When $g(c) = 0$ but $f(c) \neq 0$, the limit does not exist.
- When $g(c) = 0$ and $f(c) = 0$, the limit may or may not exist and can take any value.

In the first behavior above, when $g(c) \neq 0$, we can evaluate $\lim_{x \rightarrow c} \frac{f(x)}{g(x)}$ by using Theorem 1.3 and substituting: $\frac{f(c)}{g(c)}$.

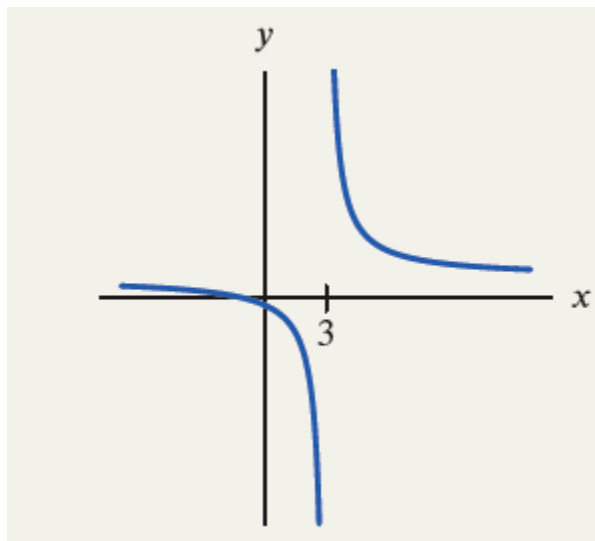
Next, consider $\lim_{x \rightarrow 3} \frac{x+1}{x-3}$. If we try to evaluate at $x = 3$ we get $\frac{4}{0}$, which is undefined.

Partner up

Let's explore this further by graphing $\frac{x+1}{x-3}$.

What happens as $x \rightarrow 3^-$ and as $x \rightarrow 3^+$?

And it's helpful for the graph, but not for the limit, but consider what happens as $x \rightarrow -\infty$ and as $x \rightarrow \infty$?



So, the limit as x approaches 3 does not exist, DNE. This is an example of the second behavior above, where $f(c) \neq 0$ but $g(c) = 0$.

When both $f(c) = 0$ and $g(c) = 0$, we have a limit of the form $\frac{0}{0}$. The limit might, or might not exist.

$$\lim_{x \rightarrow 3} \frac{x^2 - x - 6}{x - 3}.$$

Solution

If we try to evaluate at $x = 3$, we get $0/0$, which is undefined. [Figure 1.131](#) suggests that as x approaches 3, the function gets close to 5, which suggests the limit is 5.

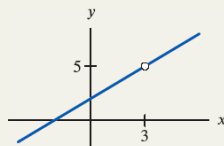


Figure 1.131: Graph of $y = (x^2 - x - 6)/(x - 3)$ is the same as the graph of $y = x + 2$ except at $x = 3$

This limit is similar to the one we saw in [Example 4](#) of [Section 1.7](#), so we check it algebraically using a similar method. Since the numerator factors as $x^2 - x - 6 = (x - 3)(x + 2)$ and $x \neq 3$ in the limit, we can cancel the common factor $x - 3$. We have:

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{x^2 - x - 6}{x - 3} &= \lim_{x \rightarrow 3} \frac{(x - 3)(x + 2)}{x - 3} && \text{Factoring the numerator} \\ &= \lim_{x \rightarrow 3} (x + 2) && \text{Canceling } (x - 3) \text{ since } x \neq 3 \\ &= 3 + 2 = 5 && \text{Substituting } x = 3 \text{ since } x + 2 \text{ is continuous} \end{aligned}$$

Even though $f(x) = x^2 - x - 6$ and $g(x) = x - 3$ approach 0 as x approaches 3, the limit in [Example 2](#) exists and is equal to 5 because the values of $f(x)$ are approximately 5 times the value of $g(x)$ near $x = 3$.

So, for the third behavior from the beginning of class, when we have a limit of the form $\frac{0}{0}$, we then have three strategies.

The first is factoring, seen in the example above.

Methods for limits as $x \rightarrow a$:

- Factor and cancel
- Multiply by a conjugate (to get rid of square roots)
- Combine fractions using common denominators

The second strategy is multiplying by a conjugate.

For example, evaluate:

$$\lim_{x \rightarrow 4} \frac{x - 4}{\sqrt{x} - 2}$$

Note that the conjugate of $\sqrt{x} - 2$ is $\sqrt{x} + 2$. What happens when $\sqrt{x} - 2$ is multiplied by $\sqrt{x} + 2$?

$(\sqrt{x} - 2)(\sqrt{x} + 2) = x - 2\sqrt{x} + 2\sqrt{x} - 4 = x - 4$. The radicals cancel out, which can simplify the limit calculation.

$$\begin{aligned} \lim_{x \rightarrow 4} \frac{x - 4}{\sqrt{x} - 2} &= \lim_{x \rightarrow 4} \left(\frac{x - 4}{\sqrt{x} - 2} \right) \left(\frac{\sqrt{x} + 2}{\sqrt{x} + 2} \right) && \text{Multiplying by 1 does not change the limit} \\ &= \lim_{x \rightarrow 4} \frac{(x - 4)(\sqrt{x} + 2)}{(\sqrt{x} - 2)(\sqrt{x} + 2)} \\ &= \lim_{x \rightarrow 4} \frac{(x - 4)(\sqrt{x} + 2)}{x - 4} && \text{Expanding the denominator} \\ &= \lim_{x \rightarrow 4} (\sqrt{x} + 2) && \text{Canceling } (x - 4) \text{ since } x \neq 4 \\ &= \sqrt{4} + 2 = 4. && \text{Substituting } x = 4 \text{ since } \sqrt{x} + 2 \text{ is continuous} \end{aligned}$$

Finally, we can combine fractions using common denominators. Remember that when you are doing the algebra, you need to keep the limit, until you evaluate it.

Consider $\lim_{h \rightarrow 0} \frac{\frac{1}{3+h} - \frac{1}{3}}{h}$

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{\frac{1}{3+h} - \frac{1}{3}}{h} &= \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{1(3)}{(3+h)(3)} - \frac{1(3+h)}{3(3+h)} \right) = \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{3 - (3+h)}{(3+h)(3)} \right) \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{-h}{(3+h)(3)} \right) = \lim_{h \rightarrow 0} \frac{-1}{(3+h)(3)} = \frac{-1}{(3+0)3} = -\frac{1}{9} \end{aligned}$$

So far, we've considered limits as $x \rightarrow c$, a constant.

We conclude by considering limits as $x \rightarrow \infty$ (or $x \rightarrow -\infty$). What is:

$$\lim_{y \rightarrow \infty} \frac{3 + 4y}{y + 2}.$$

Here we have a limit of the form $\frac{\infty}{\infty}$. The numerator and denominator both grow arbitrarily large. So we can divide by the highest power to see that the limit does exist.

$$\begin{aligned} \lim_{y \rightarrow \infty} \frac{3 + 4y}{y + 2} &= \lim_{y \rightarrow \infty} \frac{3/y + 4}{1 + 2/y} && \text{Dividing numerator and denominator by } y \\ &= \frac{0 + 4}{1 + 0} = 4 && \text{Since } 1/y \rightarrow 0 \text{ as } y \rightarrow \infty \end{aligned}$$

Note that $\frac{1}{y} \rightarrow 0$ as $y \rightarrow \infty$.